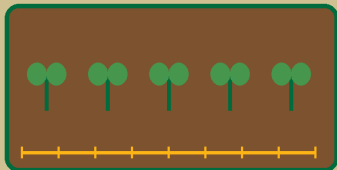


Measure Soil Moisture



MEASURE SOIL MOISTURE

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

PA-281 • Measure Soil Moisture

INVESTIGATION

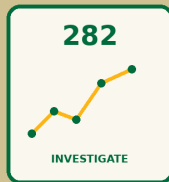
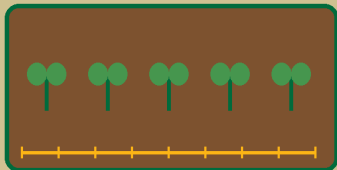
Use the appropriate teacher-approved instrument. Take measurements consistently, include units, and record the result.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Observe Infiltration



OBSERVE INFILTRATION

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

INVESTIGATION

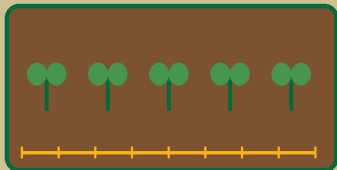
Carry out the investigation using a consistent method. Record observations or measurements and explain what the evidence indicates.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Identify Runoff



IDENTIFY RUNOFF

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

INVESTIGATION

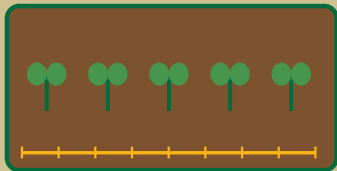
Inspect the system carefully, identify the requested feature or variable, and explain why it matters to the investigation.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Estimate Irrigation Volume



ESTIMATE IRRIGATION VOLUME

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

INVESTIGATION

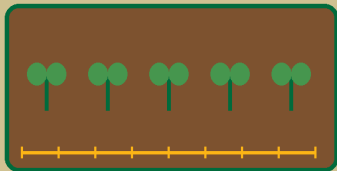
Carry out the investigation using a consistent method. Record observations or measurements and explain what the evidence indicates.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Measure Flow Rate



MEASURE FLOW RATE

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

INVESTIGATION

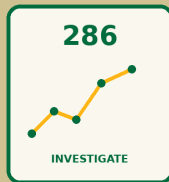
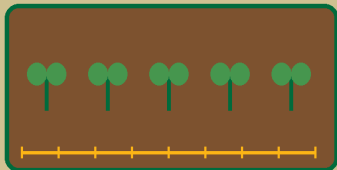
Use the appropriate teacher-approved instrument. Take measurements consistently, include units, and record the result.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Calculate Water Applied



CALCULATE WATER APPLIED

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

PA-286 • Calculate Water Applied

INVESTIGATION

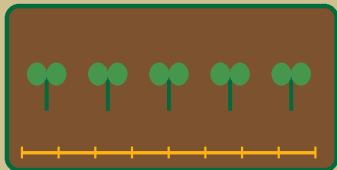
Collect or use the required measurements, show the calculation clearly, include units, and check whether the result is reasonable.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Compare Mulched and Bare Soil



COMPARE MULCHED AND BARE SOIL

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

PA-287 • Compare Mulched and Bare Soil

INVESTIGATION

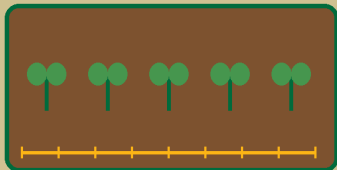
Use the same measurement or observation method for both cases.
Record the difference and identify evidence that explains it.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Create an Irrigation Schedule



CREATE AN IRRIGATION SCHEDULE

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

PA-288 • Create an Irrigation Schedule

INVESTIGATION

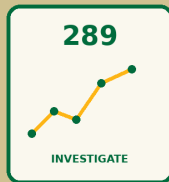
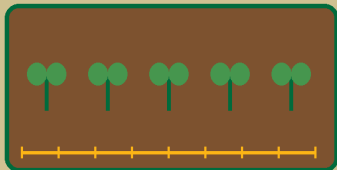
Develop a practical plan using the stated garden constraints. Label key dimensions, sequence, variables, or decision rules.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Calculate Water Productivity



CALCULATE WATER PRODUCTIVITY

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

PA-289 • Calculate Water Productivity

INVESTIGATION

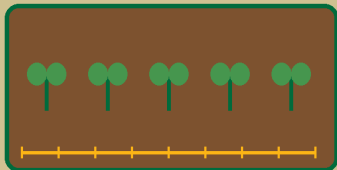
Collect or use the required measurements, show the calculation clearly, include units, and check whether the result is reasonable.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN

Make a Water-Smart Decision



MAKE A WATER-SMART DECISION

MEASURE • INVESTIGATE • ANALYZE

OBSERVE → MEASURE → INVESTIGATE → ANALYZE → EXPLAIN

INVESTIGATION

Carry out the investigation using a consistent method. Record observations or measurements and explain what the evidence indicates.

ANALYZE & EXPLAIN

What does your evidence show, and what is one reasonable explanation or improvement?

QUESTION → MEASURE → RECORD → COMPARE → EXPLAIN